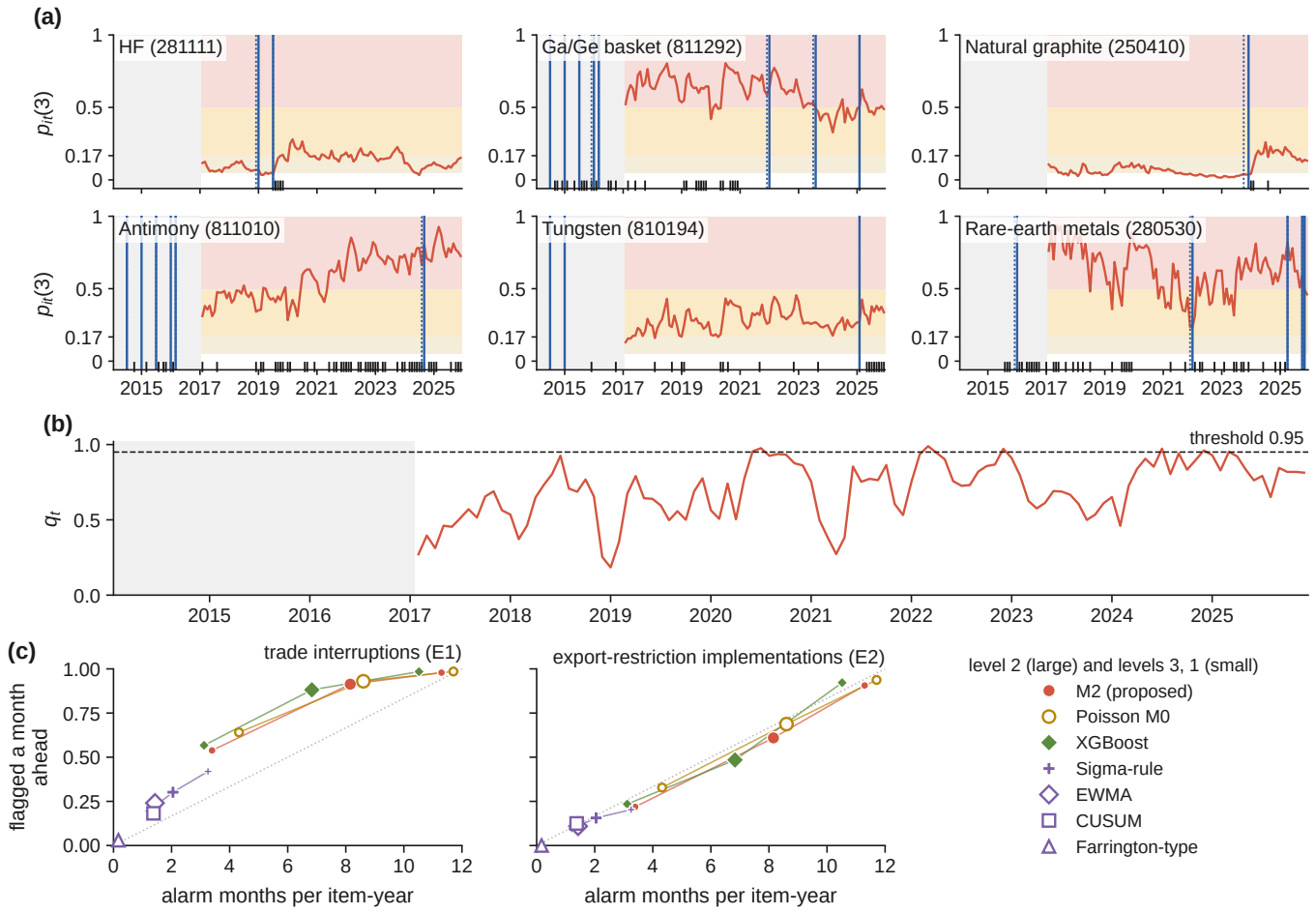
(a)
HF (281111)
Ga/Ge basket (811292)
Natural graphite (250410)
Antimony (811010)
Tungsten (810194)
Rare-earth metals (280530)
$p_{it}(3)$
1
0.5
0.17
0
2015
2017
2019
2021
2023
2025
(b)
q_t
threshold 0.95
1.0
0.5
0.0
2015
2016
2017
2018
2019
2020
2021
2022
2023
2024
2025
(c)
trade interruptions (E1)
export-restriction implementations (E2)
flagged a month ahead
alarm months per item-year
0.00
0.25
0.50
0.75
1.00
0
2
4
6
8
10
12
level 2 (large) and levels 3, 1 (small)
M2 (proposed)
Poisson M0
XGBoost
Sigma-rule
EWMA
CUSUM
Farrington-type